Polycrystal diffusion

Part A. Temperature dependence of the diffusion coefficient (2.6 points)

A1 (0.2pt)

$v =$

$D(T) \propto$

A2 (0.4pt)

$\Delta r_{\mathrm{sq}} =$

A3 (2.0pt)

T	t	N	Δr_{sq}	D

$E_g =$

Accuracy:

Part B. Effective diffusion coefficient (2.4 points)

B1 (0.4pt)

$\alpha =$

B2 (2.0pt)

t	D_{gb}/D	δ	α

$\alpha =$

Part C. Type B kinetics (5.0 points)

C1 (0.4pt)

C2 (4.0pt)

t	D_{gb}/D	δ	$\ln \bar{c}$

$\xi =$ $\eta =$

$\zeta =$

C3 (0.6pt)

$C =$

Theory

ISPhO

A2-1

English

Muon anomalous magnetic moment

Part A. Dynamics in magnetic field (4.5 points)

A1 (1.0pt)

(Formula) $\Delta v =$ (Value) $\Delta v =$

A2 (0.5pt)

(Formula) $E_k =$ (Value) $E_k =$

A3 (1.0pt)

(Formula) $R =$ (Value) $R =$

(Formula) $T =$ (Value) $T =$

A4 (0.8pt)

(Formula) $N =$ (Value) $N =$

A5 (0.7pt)

$d\vec{s}/dt =$

A6 (0.5pt)

$\omega_z =$

Part B. Thomas precession (3.5 points)

B1 (0.5pt)

$v_{Cx} =$ $v_{Cy} =$

B2 (0.7pt)

$v''_{x''} =$ $v''_{y''} =$

B3 (0.6pt)

B4 (1.0pt)

$\omega =$

B5 (0.7pt)

$\omega_{T,z} =$

Part C. Magnetic moment precession (2.0 points)

C1 (0.7pt)

$\omega_s =$

C2 (0.5pt)

$\omega_a =$

C3 (0.8pt)

(Formula) $a_\mu =$ (Value) $a_\mu =$

Theory

ISPhO

A3-1

English

Frequency measurements with ring resonators

Part A. Field profile in fiber optics (2.5 points)

A1 (0.3pt)

$C_1 =$ $C_2 =$

A2 (0.5pt)

$k_{\text{in}} =$ $k_{\text{out}} =$

A3 (0.3pt)

A4 (0.3pt)

$B =$

A5 (0.3pt)

A6 (0.3pt)

A7 (0.3pt)

Equation:

A8 (0.2pt)

Equation:

Part B. Fiber resonator transmittance (3.0 points)

B1 (0.2pt)

Equation:

B2 (0.3pt)

$B_{\text{in}}(t) =$

B3 (0.3pt)

$B_{\text{in}}(t) =$

B4 (0.5pt)

$B_{\text{in}}(t) =$

B5 (0.5pt)

$N_2 =$

B6 (0.6pt)
Plot:



B7 (0.6pt)

$Q =$

Part C. Calibration of the Mach-Zehnder Interferometer (2.0 points)

C1 (0.5pt)
Plot:



C2 (1.0pt)
Plot:



C3 (0.5pt)

$\Delta\omega_{\mathrm{MZI}}/\omega_{\mathrm{MZI}} =$

Part D. Cubic nonlinearity and solitons (2.5 points)

D1 (0.1pt)

$v_g =$

D2 (0.6pt)

D3 (0.5pt)

$D_1 =$ $D_2 =$

D4 (0.3pt)

Condition on D_2:

D5 (0.6pt)
Plot:



D6 (0.2pt)

$\Delta\omega =$

D7 (0.2pt)

$\tau_s =$